

# Solving Trigonometric Equations on Restricted Intervals

## Answer Key

Precalculus    Trigonometric Identities and Equations

### Quick Answers

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|---------------------------------------------------|-------------------------------------------------------------------|
| 1. $\frac{\pi}{6}, \frac{5\pi}{6}$                | 6. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$                            |
| 2. $\frac{5\pi}{6}, \frac{7\pi}{6}$               | 7. 13                                                             |
| 3. $-\frac{\pi}{4}, \frac{3\pi}{4}$               | 8. 10, 20                                                         |
| 4. $1 - 2\sin^2(x)$                               | 9. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ |
| 5. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$ | 10. 90, 210, 330                                                  |
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**1.** Solve  $2\sin x - 1 = 0$  on  $[0, 2\pi)$ .

**Solution.** Isolate the trig function:  $\sin x = \frac{1}{2}$ . The reference angle is  $\frac{\pi}{6}$ , and sine is positive in quadrants I and II, so the two solutions in one revolution are  $\frac{\pi}{6}$  and  $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$ .  $x = \frac{\pi}{6}, \frac{5\pi}{6}$

*Count check:* the graph crosses  $y = \frac{1}{2}$  twice on  $[0, 2\pi)$ , matching the two answers.

**2.** Solve  $2\cos x + \sqrt{3} = 0$  on  $[0, 2\pi)$ .

**Solution.**  $\cos x = -\frac{\sqrt{3}}{2}$ . The reference angle is  $\frac{\pi}{6}$ ; cosine is negative in quadrants II and III, giving  $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$  and  $\pi + \frac{\pi}{6} = \frac{7\pi}{6}$ .  $x = \frac{5\pi}{6}, \frac{7\pi}{6}$

**3.** Solve  $\tan x + 1 = 0$  on  $[-\pi, \pi)$ .

**Solution.**  $\tan x = -1$ , reference angle  $\frac{\pi}{4}$ . Tangent has period  $\pi$ , so its solutions are spaced  $\pi$  apart:  $x = -\frac{\pi}{4} + k\pi$ . The members of that family lying in  $[-\pi, \pi)$  are  $-\frac{\pi}{4}$  (from  $k = 0$ ) and  $-\frac{\pi}{4} + \pi = \frac{3\pi}{4}$  (from  $k = 1$ ).  $x = -\frac{\pi}{4}, \frac{3\pi}{4}$  *Common error:* using period  $2\pi$  and reporting only one solution, or listing  $\frac{7\pi}{4}$ , which lies outside this interval.

**4.** Verify  $\cos 2x = 1 - 2\sin^2 x$ .

**Solution.** Start from the difference form of the double-angle identity and replace  $\cos^2 x$  using the Pythagorean identity:

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= (1 - \sin^2 x) - \sin^2 x \\ &= 1 - 2\sin^2 x\end{aligned}$$

$\cos 2x = 1 - 2\sin^2 x$  This is the form to reach for whenever an equation mixes  $\cos 2x$  with  $\sin x$ : it makes every term a function of the same angle.

**5.** Solve  $\cos 2x + 3 \sin x = 2$  on  $[0, 2\pi)$ .

**Solution.** Substitute the identity from problem 4 so that every term is a function of  $\sin x$ :

$$\begin{aligned}(1 - 2 \sin^2 x) + 3 \sin x &= 2 \\ -2 \sin^2 x + 3 \sin x - 1 &= 0 \\ 2 \sin^2 x - 3 \sin x + 1 &= 0 \\ (2 \sin x - 1)(\sin x - 1) &= 0\end{aligned}$$

So  $\sin x = \frac{1}{2}$ , giving  $x = \frac{\pi}{6}$  and  $\frac{5\pi}{6}$ ; or  $\sin x = 1$ , giving  $x = \frac{\pi}{2}$ .  $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

**6.** Solve  $2 \cos^2 x - \cos x - 1 = 0$  on  $[0, 2\pi)$ .

**Solution.** Treat  $\cos x$  as the variable and factor:

$$(2 \cos x + 1)(\cos x - 1) = 0$$

$\cos x = -\frac{1}{2}$  gives the quadrant II and III angles  $\frac{2\pi}{3}$  and  $\frac{4\pi}{3}$ ;  $\cos x = 1$  gives  $x = 0$  (and  $2\pi$ , which is excluded because the interval is half-open).  $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$  *Grading note:* an answer that includes  $2\pi$  has misread the interval; an answer missing  $x = 0$  has divided by  $(\cos x - 1)$  instead of factoring.

**7.** Ferris wheel height at  $t = 10$  seconds,  $h(t) = 9 - 8 \cos(\pi t/15)$ .

**Solution.**  $\frac{\pi(10)}{15} = \frac{2\pi}{3}$ , and  $\cos \frac{2\pi}{3} = -\frac{1}{2}$ , so  $h(10) = 9 - 8(-\frac{1}{2}) = 9 + 4$ . 13 m *Sanity check:* the model has centre height 9 m and radius 8 m, so every value must lie between 1 m and 17 m; 13 m does.

**8.** Find every  $t \in [0, 30)$  seconds with  $h(t) = 13$  metres.

**Solution.** Set the model equal to the target and isolate the cosine:

$$\begin{aligned}9 - 8 \cos\left(\frac{\pi t}{15}\right) &= 13 \\ \cos\left(\frac{\pi t}{15}\right) &= -\frac{1}{2}\end{aligned}$$

Let  $u = \frac{\pi t}{15}$ . As  $t$  runs over  $[0, 30)$ ,  $u$  runs over  $[0, 2\pi)$  — exactly one revolution — so  $u = \frac{2\pi}{3}$  or  $u = \frac{4\pi}{3}$ . Converting back,  $t = \frac{15u}{\pi}$  gives  $t = 10$  and  $t = 20$ .  $t = 10 \text{ s}, 20 \text{ s}$  The two times are symmetric about  $t = 15$  s, the top of the wheel — the rider passes 13 m once going up and once coming down.

**9.** Solve  $\sin 2x = \cos x$  on  $[0, 2\pi)$ .

**Solution.** Rewrite the double angle, move everything to one side, and factor — never divide by  $\cos x$ , which would delete the solutions where  $\cos x = 0$ :

$$\begin{aligned} 2 \sin x \cos x - \cos x &= 0 \\ \cos x (2 \sin x - 1) &= 0 \end{aligned}$$

$\cos x = 0$  gives  $x = \frac{\pi}{2}$  and  $\frac{3\pi}{2}$ ;  $\sin x = \frac{1}{2}$  gives  $x = \frac{\pi}{6}$  and  $\frac{5\pi}{6}$ .  $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$  *Grading note:* exactly the two answers a student loses by dividing through by  $\cos x$  are  $\frac{\pi}{2}$  and  $\frac{3\pi}{2}$ .

**10.** Solve  $2 \sin^2 \theta - \sin \theta - 1 = 0$  on  $[0^\circ, 360^\circ)$ .

**Solution.** Factor the quadratic in  $\sin \theta$ :

$$(2 \sin \theta + 1)(\sin \theta - 1) = 0$$

$\sin \theta = 1$  gives  $\theta = 90^\circ$ .  $\sin \theta = -\frac{1}{2}$  has reference angle  $30^\circ$  and is negative in quadrants III and IV, giving  $180^\circ + 30^\circ = 210^\circ$  and  $360^\circ - 30^\circ = 330^\circ$ .  $\theta = 90^\circ, 210^\circ, 330^\circ$  *Grading note:*  $\sin \theta = 1$  contributes only one solution per revolution because 1 is the maximum of sine — a student who reports two angles there has applied the quadrant rule without checking the extreme value.